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Project Proposal 01 — Les Misérables Character Co-occurrence Network

Analisi e Visualizzazione delle Reti Complesse — Network Science module, final project.

Abstract

Donald E. Knuth's co-occurrence network of Victor Hugo's *Les Misérables* is the smallest dataset in this folder and one of the most rewarding. It has **77 nodes and 254 edges**. Each node is a named character. Each edge connects two characters who appear together in at least one chapter. Because every number you compute can be checked against the plot of the novel, the dataset is an excellent training ground for interpretation.

The project produces a structural portrait of the book. It asks: who are the structural protagonists? How do the algorithmic communities map onto the plot? How far does the topology deviate from chance? The result is a compact, readable report supported by a small number of rigorous experiments and one carefully designed figure.

1. Introduction and motivation

A convincing network-science project does not need millions of edges. This dataset shows why: its value is its transparency. When Louvain finds five communities, one of them is the Thénardiers, one is the students of the ABC, one is the convent — and you can say so in a sentence. When betweenness centrality promotes a character that degree centrality would not, you can open the novel and explain why. This back-and-forth between metric and meaning is the core skill the course tries to teach, and *Les Misérables* is the cleanest dataset on which to practise it.

Choosing this project means accepting a specific trade-off. The analytical work is light — every algorithm in the syllabus runs in seconds — but the interpretive work is heavy. The report has to say something about the novel, not just about its graph. In exchange, the figures are beautiful, the narrative writes itself, and the oral exam becomes a pleasant conversation.

A useful reference point: Newman's 2004 "fast algorithm" paper, where many students first meet modularity, runs its greedy community-detection method on exactly this network and reports **11 communities at modularity Q ≈ 0.54**. You can use this as an anchor when evaluating your own partition.

2. The dataset

2.1 Source and provenance

The network was compiled by Donald E. Knuth for *The Stanford GraphBase: A Platform for Combinatorial Computing* (Addison-Wesley, 1993), file `jean.dat`. NetworkX ships it as `nx.les_miserables_graph()`, so no download is required. A reference copy is also hosted on KONECT.

2.2 Structure and semantics

- **77 nodes**, **254 undirected, weighted edges**.

Figure 1: Hero figure: Les Misérables co-occurrence network, coloured by Louvain community, sized by weighted betweenness centrality. The algorithmic partition recovers narrative groups that any reader of the novel recognises immediately.

- A node is a named character.
- An edge between two characters means they co-appear in at least one chapter; the weight is the number of shared chapters.
- The graph is already connected.

2.3 Caveats

- Chapters have different lengths, so the edge weight is a noisy proxy for tie strength.
- Any analysis with a `weight=` argument gives different answers on the weighted and unweighted versions — pick one per experiment and state it clearly.
- With only 77 nodes, any statistical claim must be hedged. A power-law fit here is a teaching exercise, not a scientific finding.

3. Research questions

Pick **two or three** of the following. Depth beats breadth. Each question is resolved with tools from a specific block of the course.

1. **Who are the protagonists, structurally?** Hugo's novel has obvious protagonists — Valjean, then Javert, Cosette, Marius, Gavroche. But "central" means different things in network terms. Is Valjean central because he has the most connections (degree)? Because he sits on the shortest paths between everyone else (betweenness)? Because he is connected to other central characters (eigenvector)? These are distinct claims. The five measures of **Lecture 6** each capture one notion of importance. On this dataset the interesting findings come from the places where the measures *disagree*: a character with high betweenness but moderate degree is a *narrative broker* — a character whose importance comes from being the only link between otherwise disconnected parts of the story. *Tools:* the five centralities of **Lecture 6**; pairwise Spearman rank correlations; top-k tables per measure; qualitative interpretation of rank disagreements against the novel.

2. **Does the graph's community structure match the novel's plot?** Any reader of *Les Misérables* carries an intuitive social partition of the book — the bishop's household, the Thénardier circle, the students of the ABC, the convent, the police. The question is whether a community-detection algorithm, which has never read the novel, recovers this partition from the co-occurrence graph alone. If it does, the result shows that plot structure is encoded in social topology. If it fails, the mismatch is itself informative. Use at least two algorithms from the scalable community-detection family of **Lectures 12–15** (for example Louvain and label propagation). Evaluate each partition against a hand-annotated "role" label for the 15–20 main characters. Newman (2004) reports $Q \approx 0.54$ with 11 communities on this same graph — a useful reference. *Tools:* modularity (**Lecture 13**); Louvain under multiple seeds; pairwise NMI / ARI as a stability check (**Lecture 15**); confusion table between algorithmic partition and hand-annotated roles.

3. **Is the network assortative by role?** Homophily — the tendency of similar nodes to connect — is the subject of **Lecture 5**. Most datasets measure it on flat demographic labels. On *Les Misérables* the labels are narratively meaningful: clergy, student, police, criminal, bourgeois, orphan. Do heroes mostly talk to heroes? Do the criminals cluster? Are the protagonists the cross-category bridges? A moderately positive assortativity is the expected answer and makes a quantitative point: *Les Misérables* is a story about categories crossing. *Tools:* degree assortativity; attribute assortativity on the annotated roles; the mixing matrix as a visual summary.

4. **What shape has each protagonist's world?** The novel contains several archetypes of protagonism, and the ego-network view makes them concrete:

   - **Valjean's ego network** is dense and multi-community — the shape of a character whose arc threads every sub-plot.
   - **Javert's** is narrow and institutional — the shape of a functionary.
   - **Marius's** is bridging — he is the node through which the ABC students meet Cosette and, through her, Valjean.

   The question is whether these three archetypes — hub, institutional, broker — show up quantitatively in the ego subgraphs. This is **Lecture 4** applied directly. *Tools:* ego-network extraction; clustering coefficient of

each ego subgraph; per-edge neighbourhood-overlap distribution; count of local bridges; Burt's effective size and constraint.

5. **Is the macrostructure distinguishable from chance?** Raw numbers need a reference. A density of 0.09 is neither large nor small by itself — it is large compared to one baseline and small compared to another. The null models of **Lecture 8** supply those baselines: an Erdős–Rényi graph of matched density, and a configuration model preserving the degree sequence. The real graph should deviate sharply from both on clustering and modularity, while being almost indistinguishable on average path length. This is the classic "structure above chance" signature. *Tools:* Erdős–Rényi and configuration-model graphs averaged over ≥ 50 realisations; means and standard deviations compared against the observed values.

6. *(optional)* **Does centrality translate into spreading power?** A character who is structurally central should also be a good carrier of disease or rumour. This is a *claim*, not a tautology, and it can be tested with an SI or SIR simulation (**Lecture 22**) seeded at Valjean and, separately, at a peripheral character such as Mabeuf. If structural centrality implies dynamical centrality — a connection the course develops in **Lectures 22 and 23** — the curve seeded at Valjean should rise faster and reach a higher plateau.

4. Expected findings

A well-executed project should reach, at minimum, the following results:

- A Louvain or Leiden partition at modularity Q in the **0.50–0.60 range**, comparable to the **Q ≈ 0.54** reported by Newman (2004) on this same graph. The largest communities should correspond to recognisable narrative groups. Role-purity should exceed 0.70 on at least three of them.
- A centrality table dominated by **Valjean, Gavroche, Myriel, and Marius**, with pairwise Spearman rank correlations in the **0.55–0.85 range** and at least one clear "broker" outlier (high betweenness, moderate degree).
- Attribute assortativity on hand-annotated roles in the **+0.2 to +0.4 range** — weakly homophilous, consistent with a novel about categories crossing.
- A null-model comparison in which observed clustering exceeds the configuration-model expectation by a factor of at least two, and observed modularity by a comparable margin — while average path length is almost indistinguishable from the null.

A report that falls substantially short of these numbers is likely hiding a methodological bug.

5. Visualization plan

Visual design is the group's responsibility — the dataviz module covers the *how*. This section only states the *what*: for each finding the project claims, a visualisation should carry the corresponding message.

- **Structural portrait of the graph.** The report's *hero figure*: a single, carefully designed view that communicates the overall shape of the novel's social network — centrality, communities, bridges.
- **"Which character is important, and by which definition?"** A visualisation that compares several centrality measures at the same time and exposes the characters on which the measures disagree.
- **"Do algorithmic communities match the novel's groups?"** A visualisation that places the algorithmic partition next to the hand-annotated roles so the reader can judge the correspondence at a glance.
- **"What does an individual protagonist's world look like?"** A comparative view of two or three selected ego-networks, drawn at the same scale, so different archetypes of protagonism are visible next to each other.
- **"Is this graph distinguishable from chance?"** A visualisation that places the observed value of each key metric against the distribution produced by a null model.

An interactive version of the hero figure — feasible here because every character can be hovered — is optional but rewarded. It should use interaction for something the static figure cannot do: browse a character's neighbourhood, toggle between views, filter by community.

6. Possible extensions

- **Text-derived network.** Parse the Project Gutenberg text of the novel and build a sentence-level co-occurrence network. Compare it against Knuth's chapter-level graph.

- **Signed variants.** Hand-annotate a dozen edges as antagonistic versus friendly and apply the structural-balance theory of **Lecture 5** to the resulting signed sub-network.
- **Cross-novel comparison.** Apply the same pipeline to a published co-occurrence network of another novel (Anna Karenina, Moby-Dick, Shakespeare) for a structural comparison.
- **Generative fits.** Try to reproduce the degree and clustering profile with a Watts–Strogatz graph or a stochastic block model fitted on the Louvain partition (**Lecture 15**).

7. Common pitfalls

- **Statistical confidence** is the subtle trap on a small graph. A single NMI between two partitions is not a result — always average over multiple seeds.
- **Power-law claims on 77 data points** should be avoided, or explicitly hedged.
- **Always state** whether a given number was computed on the weighted or unweighted version of the graph.
- **Do not redraw the hero figure twenty times.** Invest the time once, in tuning a small number of design decisions.

References

- Knuth, D. E. (1993). *The Stanford GraphBase: A Platform for Combinatorial Computing.* Addison-Wesley.
- Newman, M. E. J. (2004). *Fast algorithm for detecting community structure in networks.* Physical Review E 69, 066133 — the reference Louvain-predecessor analysis on this exact dataset.
- Blondel, V. D., Guillaume, J.-L., Lambiotte, R., and Lefebvre, E. (2008). *Fast unfolding of communities in large networks.* Journal of Statistical Mechanics.
- Traag, V. A., Waltman, L., and van Eck, N. J. (2019). *From Louvain to Leiden.* Scientific Reports.

- Course syllabus and lecture slides: `netsci/ns_syllabus.md`, `netsci/pdfs/`.